Algorithms, 2026 Fall, Homework 2
(Due: Oct 1)

September 22, 2026

Required Readings: Please do the following if you have not done so.

1. Chapter 3.3 Stack and Queue.
2. Chapter 2.2 Priority queue.
3. If you do not know Markov inequality, Chebyshev inequality, Union bound and Chernoff bound, you should quickly find some references and learn them. They are basic but very important in randomized algorithms and other domains where probabilities appear. You can read the first 3 pages in http://www.cs.cmu.edu/afs/cs.cmu.edu/academic/class/15859-f04/www/scribes/lec9.pdf You can also read the textbook Chapter 13.9 for Chernoff Bounds and page 772 for Union Bound. (You need to finish reading this within the first three weeks)

You can discuss homeworks with your classmates. But you will have to write down the solution on your own. You need to acknowledge other students who help you in the homework. If you read some other source that is helpful, you should list all of them.

Problems marked with (@) can be skipped by those who have extensive experience in OI

Problem 1 :(@) KT book, pp 107, Problem 2

Problem 2 :(@) Write down a pseudocode implementation of Kruskal's algorithm using the Union-Find data structure. (you can assume the union-find data structure is ready and you can use it directly – by calling union and find operations) Your algorithm should run in time $O(n \log n)$.

Problem 3 : We say that a vertex v in an undirected graph G is an *articulation point* if its removal increases the number of connected components of G.

(1) Try to detect whether v is an articulation point using DFS.

(2) Try to find all articulation point using DFS (your algorithm should run in $O(|V| + |E|)$ time). (hint: You can show the following characterization of articulation point. Suppose you are visiting node v in DFS: (1) If v is the root of the DFS tree and it has at least 2 children subgraphs disconnected from each other, then v is an articulation point. (2) If there is no way to reach a node u with strictly smaller discovery time than that of v following the DFS traversal, then v is an articulation point.)

Problem 4 : KT book, pp 110, Problem 9

Problem 5 : KT book, pp 193, Problem 12

Problem 6 :(@) (KT pp.200, problem 22) We have assumed in the class that all edge weights are distinct (for the minimum spanning tree problem). If the edges cost in a graph are not all distinct, there might be many minimum spanning trees. Hence, our proofs (for Prim algorithm and Kruskal's algorithm) may require a bit modification). (1) Formally prove both Prim algorithm and Kruskal's algorithm still work even not all weights are distinct. (2) Suppose you have a spanning tree T such that each edge in T belongs to some minimum spanning tree. Can we conclude T itself must be a minimum-cost spanning tree? Give a proof or a counterexample.

Problem 7 : KT book, pp 198, Problem 19